

RandomForestRegressor

1. N_Estimators = 100

```
Initializing Random Forest Regressor...
Training on 100 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.8047 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 2.9056e+14
```

```
Initializing Random Forest Regressor...
Training on 1000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9644 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.2407e+14
```

```
Initializing Random Forest Regressor...
Training on 2000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9764 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.0107e+14
```

```
Initializing Random Forest Regressor...
Training on 5000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9824 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 8.7206e+13
```

```
Initializing Random Forest Regressor...
Training on 7500 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9858 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 7.8343e+13
```

```
Initializing Random Forest Regressor...
Training on 10000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9857 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 7.8627e+13
```

```
Initializing Random Forest Regressor...
Training on 15000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9911 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 6.2170e+13
```

```
Initializing Random Forest Regressor...
Training on 20000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9924 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 5.7201e+13
```

2. N_Estimators = 200

```
Initializing Random Forest Regressor...
Training on 100 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.8026 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 2.9208e+14
```

```
Initializing Random Forest Regressor...
Training on 1000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9650 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.2301e+14
```

```
Initializing Random Forest Regressor...
Training on 2000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9766 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.0059e+14
```

```
Initializing Random Forest Regressor...
Training on 5000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9824 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 8.7263e+13
```

```
Initializing Random Forest Regressor...
Training on 10000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9860 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 7.7855e+13
```

```
Initializing Random Forest Regressor...
Training on 15000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9911 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 6.2021e+13
```

```
Initializing Random Forest Regressor...
Training on 20000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9925 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 5.7105e+13
```

3. N_Estimators = 500

```
Initializing Random Forest Regressor...
Training on 100 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.8079 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 2.8819e+14
```

```
Initializing Random Forest Regressor...
Training on 1000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9646 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.2369e+14
```

```
Initializing Random Forest Regressor...
Training on 2000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9763 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 1.0119e+14
```

```
Initializing Random Forest Regressor...
Training on 5000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9823 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 8.7424e+13
```

```
Initializing Random Forest Regressor...
Training on 10000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9860 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 7.7871e+13
```

```
Initializing Random Forest Regressor...
Training on 15000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9913 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 6.1440e+13
```

```
Initializing Random Forest Regressor...
Training on 20000 sample rows...
Training complete!
Predicting test data...
-----
Random Forest R^2 Score: 0.9925 (Closer to 1.0 is better)
Root Mean Squared Error (RMSE): 5.6969e+13
```

GaussianProcessRegressor

1. N_restarts = 3

```
Initializing GPR with Matérn 5/2 kernel...
Training on 1000 sample rows. This may take a minute or two...
Training complete!
Predicting test data to evaluate accuracy...
-----
Model R^2 Score: 0.3624 (Closer to 1.0 is better)
Mean Absolute Error: 346.03% (Average prediction error)
```

```
*** Initializing GPR with Matérn 5/2 kernel...
Training on 2000 sample rows with 3 restarts.
Training complete!
Predicting test data to evaluate accuracy...
-----
Model R^2 Score: 0.6114 (Closer to 1.0 is better)
Mean Absolute Error: 152.38% (Average prediction error)
```

```
Initializing GPR with Matérn 5/2 kernel...
Training on 5000 sample rows with 3 restarts.
Training complete!
Predicting test data to evaluate accuracy...
-----
Model R^2 Score: 0.9038 (Closer to 1.0 is better)
Mean Absolute Error: 92.84% (Average prediction error)
```

2. N_restarts = 5

```
Initializing GPR with Matérn 5/2 kernel...
Training on 1000 sample rows with 5 restarts.
Training complete!
Predicting test data to evaluate accuracy...
-----
Model R^2 Score: 0.3624 (Closer to 1.0 is better)
Mean Absolute Error: 346.03% (Average prediction error)
```

```
Initializing GPR with Matérn 5/2 kernel...  
Training on 2000 sample rows with 5 restarts.  
Training complete!  
Predicting test data to evaluate accuracy...  
-----  
Model R^2 Score: 0.6114 (Closer to 1.0 is better)  
Mean Absolute Error: 152.38% (Average prediction error)
```